A Multistrategy Evolutionary Multiobjective Optimization Method for Hyperspectral Endmember Extraction

Chuanlong Ye, Fazhi He, *Member, IEEE*, Jinkun Luo, Lyuyang Tong, Xiaoxin Gao, Tongzhen Si, and Linkun Fan

***Abstract*— Hyperspectral endmember extraction (HEE) is an essential part of remote-sensing image processing. There have been recent attempts to model the HEE as a multiobjective optimization problem and apply multiobjective evolutionary algorithms to solve the problem. However, because of the large HEE search space, it is difficult for the current algorithms to achieve exploration–exploitation balance, and they easily stall prematurely. To address these issues, this article proposes a multistrategy evolutionary multiobjective method based on roulette wheel selection and the genetic algorithm (RWS-GA) for endmember extraction. This method designs two parallel algorithms corresponding to global exploration and local exploitation. In the RWS-GA, an improved NSGA-II method, adopting a novel method to sort individuals on the same front instead of the crowding distance, is proposed to divide individuals into superior and inferior subpopulations. Thereafter, different modified population update strategies are utilized for subpopulations based on characteristics. Pixels that appear more frequently in the population are considered to perform better to have a higher probability of forming an endmember set with other pixels. In addition, excellent individuals often exhibit a higher probability of including endmembers compared with inferior individuals. Considering the abovementioned opinions, roulette wheel selection is performed on the inferior subpopulation for global search. Meanwhile, the superior subpopulation is responsible for local search based on the genetic algorithm (GA). Furthermore, an offspring complement mechanism (OCM) is presented to prevent duplicate individuals from appearing in historical archives. Numerous comparative experiments show that the proposed method is superior to other endmember extraction methods in three real-world datasets.**

***Index Terms*— Evolutionary algorithms, hyperspectral endmember extraction (HEE), multiobjective optimization, NSGA-II.**

I. Introduction

HYPERSPECTRAL images have become an important and popular data source for remote sensing [1], [2], [3], [4], [5] due to the detailed spectral surface information [6]. Because of the resolution restriction, mixed pixels are inevitable in remote-sensing images [7]. In the absence of auxiliary data, hyperspectral unmixing technology is an important means for solving the mixed pixels problem and improving remote-sensing application accuracy.

The main task of hyperspectral unmixing is to identify the material (endmember) and corresponding proportion (abundance) of features contained in each pixel [8], [9]. According to the specific process, hyperspectral unmixing can be roughly divided into two major categories. The first category obtains the endmember and abundance matrices of the image simultaneously. The specific algorithms include nonnegative matrix decomposition [10], [11] and deep learning methods based on autoencoders [12], [13]. The second type of method evaluates the abundance matrix after obtaining endmember signatures; among them, sparse representation-based algorithms are typical algorithms [14], [15]. In addition, common endmember extraction algorithms include four categories, pure pixel assumption-based algorithms [16], [17], minimum volume model-based algorithms [18], spatial information-based algorithms [19], and statistical model-based algorithms [20]. However, these methods may become stuck in a local optimum and cannot extract accurate endmembers effectively for images with noise and outliers.

As evolutionary algorithms mature, scholars have attempted to apply them to the hyperspectral unmixing field [21], [22], especially for endmember extraction [23]. Under this condition that there are pure pixels in the image, the purpose of the hyperspectral endmember extraction (HEE) is to extract a pixel set from the image that represents the spectral characteristics of various ground objects. As described above, the HEE is essentially a combinatorial optimization problem, which prompts researchers to introduce evolutionary optimization methods into this field. Initially, researchers carried out single-objective optimization algorithms to extract endmembers and adopted the root-mean-square error (RMSE) between the original image and the reconstructed image [24] or the volume of the simplex composed of extracted endmembers as the objective function [25].

Particle swarm optimization (PSO) [26], [27] is a common heuristic algorithm used for image processing [28]. Omran et al. [29] proposed a combination method based on PSO and K-means to extract endmembers in remote-sensing images and took RMSE as the objective function. After that,

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Zhang et al. [24] proposed a discrete PSO (DPSO) optimization algorithm and designed a task-specific speed update strategy. Subsequently, Xu et al. [25] introduced quantum-behaved PSO (QPSO) into endmember extraction and proposed a novel mutation operator-accelerated QPSO (MOAQPSO). Du et al. [30] proposed the improved QPSO (IQPSO) and encoded the particles with the 2-D coordinate. In HEE, ant colony optimization (ACO) and artificial bee colony (ABC) are also commonly used heuristic algorithms. Due to the advantages of ACO in combinatorial optimization problems, Xu et al. [31] and Zhang et al. [32] introduced ACO into the domain of endmember extraction and viewed each endmember as a path. In addition, Yang et al. [33] adopted the ABC into the HEE to overcome the limitation of computational efficiency. Fu et al. [34] proposed a discrete algorithm incorporating the genetic algorithm (GA) and ABC for endmember extraction. In addition to the above algorithms, many other algorithms are also modified for endmember extraction, including the differential evolution (DE) algorithm [35], cuckoo algorithm [36], and firefly algorithm [37].

However, for real-world applications, the original plan may need to be adjusted depending on the situation; therefore, multiobjective optimization is more practical and significant than single-objective optimization. When noise points or outliers were extracted as endmembers, researchers found that the endmember collection is often accompanied by a considerable volume and a poor reconstruction error [38]. In other words, volume and reconstruction error are two conflicting objective functions, which conform to the premise of the multiobjective question.

Currently, multiobjective optimization algorithms are applied to the HEE to obtain trade-off solutions between two objective functions [39]. Among them, PSO still plays an important role. Liu et al. [38] proposed a best local guide strategy and designed a novel algorithm named multiobjective discrete particle swarm optimization algorithm (MODPSO) based on DPSO. Moreover, Tong et al. [40] further proposed improved MODPSO (IMODPSO), which solves the problem of parameter selection in MODPSO. Cheng et al. [41] proposed ANSGA-III based on NSGA-II for endmember extraction. A multiobjective DE algorithm based on ranking multiple mutations called $(\mu + \lambda)$-MODE was proposed by Tong et al. [42], and the result outperformed previous evolutionary optimization methods.

Drawing from the aforementioned research, it is evident that while multiobjective algorithms for endmember extraction have exhibited considerable promise in real-world scenarios, they are still in their nascent stages. The utilization of multiobjective optimization algorithms for HEE is confronted with two significant challenges. The first challenge pertains to the complexity of the combinatorial optimization problem associated with endmember extraction, which necessitates a higher level of optimization than other problems due to the vast search space. For a scene containing 10 000 pixels and three endmembers, the number of combinations already reaches a high magnitude. Due to the large search space of endmember extraction, current algorithms have difficulty balancing exploration and exploitation. The second challenge involves the need for a more comprehensive exploration of multiobjective optimization in conjunction with endmember extraction. Although endmember extraction is a discrete problem, several algorithms still employ continuous approaches, resulting in postprocessing adjustments to the obtained pixel index, thereby compromising the physical significance of the outcomes. Meanwhile, most of the current discrete optimization algorithms for HHE adopt the personal best particle and the global particle to update individuals, which easily stall prematurely.

By further exploring the combination of endmember extraction and multiobjective optimization, two ideas are proposed as follows.

Idea 1: Individuals that perform well often have a higher probability of including endmembers compared with secondary individuals.

Idea 2: Pixels that appear more frequently in a population have a higher probability of being considered endmembers.

To solve the above problems, an algorithm called roulette wheel selection and the GA (RWS-GA) based on the above two ideas is proposed. For the challenges, based on Idea 1, individuals are first divided into two populations based on their performance. To better sort the population and enhance its diversity, an improved NSGA-II is proposed, in which a new sorting method for individuals on the same front is proposed to replace the original crowding distance sorting. Afterward, excellent individuals are responsible for local exploitation, while disadvantaged individuals are responsible for global exploration. The two evolutionary strategies run in parallel to balance as much of the exploration–exploitation capability of the algorithm as possible. In this article, each pixel is assigned a specific label, and each individual contains a fixed number of pixels, that is, the number of endmembers. For local search, superior subpopulations will recombine pixels based on GAs [43] to form new individuals. In addition, according to Idea 2, the roulette selection algorithm [44] was redesigned for global search. In addition, the evaluation of the objective function is a time-consuming process. To avoid duplicate calculations caused by generating the same individuals, an offspring complement mechanism (OCM) is designed to enhance population diversity.

The rest of this article is organized as follows. Section II introduces the related work on linear mixed models and multiobjective optimization. The proposed RWS-GA method is presented in detail in Section III. In the fourth part, the proposed method is compared with other methods in actual datasets. Finally, we summarize this article.

II. Related Work

A. Linear Spectral Mixing Model

In the linear model, the spectra of pixels in the image are composed of two parts: the endmember-weighted corresponding abundance and noise. The spectral signature of a pixel is formed as follows [45]:

$$y = \sum_{i=1}^{P} \boldsymbol{a}_i \boldsymbol{e}_i + n \tag{1}$$

where y represents the spectrum of the pixel, $\boldsymbol{a}$ contains the corresponding abundance of each endmember in the pixel,

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$\boldsymbol{e}$ denotes the endmember set, and n stands for noise during imaging.

By extending the above single-pixel model to the whole image, the following equation can be obtained:

$$\mathbf{Y} = \mathbf{EA} + \mathbf{N} \tag{2}$$

where the hyperspectral image $\mathbf{Y}$ is a matrix with a size of $L \times NP$, consisting of L bands and NP pixels. The spectral library $\mathbf{E}$ contains P spectra, with each column representing an endmember signature. $\mathbf{A}$ indicates the P-by-NP abundance matrix. $\mathbf{N}$ represents the noise matrix with the same size as $\mathbf{Y}$.

There are two constraints commonly used in unmixing, namely, the abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC), which are expressed in a single pixel as follows:

$$\boldsymbol{a} \geq 0 \tag{3}$$

$$\mathbf{1}^T \boldsymbol{a} = 1. \tag{4}$$

There are four least squares methods for abundance estimation based on these two constraints. However, it has been found that ASC may make image reconstruction less accurate [46]. Herein, the nonnegative constrained least squares (NCLS) method is adopted, which only considers ANC [47].

B. Multiobjective Optimization in Endmember Extraction

When multiple conflicting objectives need to be optimized simultaneously in a problem, this problem is referred to as a multiobjective problem [48], which can be demonstrated as follows:

$$\min \boldsymbol{F}(x) = [f_1(x), f_2(x), \ldots, f_m(x)] \quad \text{s.t. } x \in \Omega \tag{5}$$

where m is the number of objective functions and x represents one candidate solution, that is, a collection of endmembers extracted from the hyperspectral image. $f_i(x)$ indicates the objective function. As mentioned in the introduction above, there are two objective functions to be considered simultaneously, including the volume inverse and reconstruction error. Ω denotes the search space, in other words, all pixel combinations in the image.

Conflicts often emerge between different objective functions, so two objective functions cannot be optimized simultaneously. Assume there is a solution x^*, and another solution x satisfies the following conditions [49]:

$$\begin{cases} f_i(x^*) \leq f_i(x), & \forall i \in \{1, 2, \ldots, m\} \\ f_j(x^*) < f_j(x), & \exists j \in \{1, 2, \ldots, m\}. \end{cases} \tag{6}$$

Then, we take the solution x^* as a Pareto solution, and the projection of Pareto solutions in the objective space is called the Pareto front (PF). The multiobjective optimization task is to obtain solutions as close to the Pareto front as possible.

The inverse of the volume and reconstruction error of candidate solution x can be computed by the following equations, respectively:

$$f_1(x) = \frac{1}{\text{volume}(E)} = \frac{(P-1)!}{\left|\det\begin{bmatrix} 1 & 1 & \cdots & 1 \\ \widetilde{x}_1 & \widetilde{x}_2 & \cdots & \widetilde{x}_P \end{bmatrix}\right|} \tag{7}$$

$$f_2(x) = \text{RMSE}(\mathbf{Y}, \widehat{\mathbf{Y}}) = \frac{1}{NP} \sum_{i=1}^{NP} \sqrt{\frac{1}{L} \left\| y_i - \widehat{y}_i \right\|_2^2} \tag{8}$$

where volume(E) represents the volume of the endmember collection corresponding to the candidate solution x. The volume is calculated after dimension reduction with principal component analysis [50], and $\widetilde{x}_i$ represents the dimensionality reduction result of the ith pixel in the candidate solution x, with the length of $P-1$. $\widehat{\mathbf{Y}}$ and $\mathbf{Y}$ indicate the original image and reconstructed image, respectively.

III. Multistrategy Optimization Method Based on RWS-GA

This section provides the details of the proposed RWS-GA method, which first divides individuals into two subpopulations based on the improved NSGA-II. Population updating relies on three strategies. In addition to parallel global exploration and local search strategies, the offspring completion mechanism also achieves population updating by mobilizing the global search strategy. The specific content of the proposed algorithm is arranged as follows.

A. Population Initialization

Population initialization is the first step in RWS-GA, and it is also essential preliminary work. In the case of an image containing NP pixels, N individuals containing P pixels need to be generated. In general, NP is much larger than the product of P and N. A random sequence of integers is generated, ranging from 1 to NP. After that, the first $P \times N$ numbers are selected and assigned to N individuals.

B. Improved NSGA-II

NSGA-II [51], [52] is a classical nondominated sorting method and includes two steps. First, each individual is determined to belong to a front based on fast nondominated sorting. Then, the crowding distance of each individual on the same front is calculated, and the specific formula is shown in the following equation:

$$[i]_{\text{distance}} = \sum_{k=1}^{m} \frac{f[i+1]_k - f[i-1]_k}{f_k^{\max} - f_k^{\min}} \tag{9}$$

where $[i+1]_k$ and $[i-1]_k$ are the adjacent individuals of individual i on the kth objective. $f_k^{\max}$ and $f_k^{\min}$ are the maximum and minimum values of the kth objective function. According to NSGA-II, individuals with larger crowding distances are better than those in the same front. Eventually, for the next iteration, individuals on lower fronts are selected preferentially, followed by individuals with better crowding distances. From formula (9), it is evident that NSGA-II only considers the distance between each individual and its neighboring individuals, which has limitations and can easily

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lead to uneven distribution of the selected individuals at the same front. In other words, individuals with worse crowding distance may be more suitable to demonstrate the diversity of the front.

Based on the above problems, an improved NSGA-II is proposed, and its main innovation is to propose a new individual sorting strategy for individuals on the same front instead of crowding distance sorting. After all the individuals are assigned to the corresponding fronts, for each front, the maximum and minimum values of each objective function are extracted and recorded. Then, the relative distance between all individuals is calculated according to the following equation:

$$\text{Distance}(i, j) = \sum_{k=1}^{m} \frac{|f_k(i) - f_k(j)|}{|f_k^{\max} - f_k^{\min}|} \tag{10}$$

where $\text{Distance}(i, j)$ represents the relative distance between individuals i and j. $f_k(i)$ signifies the value of individual i on the kth objective, and $f_k^{\max}$ and $f_k^{\min}$ implicate the maximum and minimum values of all individuals on the same front on the kth objective, respectively.

After obtaining the distance matrix distance, vector V_{out} and empty vector V_{in} are generated, where V_{out} records individuals on the front. The individuals corresponding to the minimum value of each objective function are moved from V_{out} into V_{in}. Then, the submatrix of the distance matrix is extracted, which records the distance between the individuals in V_{out} and the individuals in V_{in}. The minimum distance of each individual in V_{out} to all individuals in V_{in} is recorded. Hereafter, the individual in V_{out} corresponding to the maximum value is moved into V_{in}. The sorting of individuals is determined by the order of individuals entering the vector V_{in}. The pseudocode for this algorithm is given in Algorithm 1.

Here is an example for Algorithm 1. There are eight points on the same front, including A (0.18, 0.82), B (0.95, 0.15), C (0.61, 0.40), D (0.55, 0.50), E (0.46, 0.53), F (0.42, 0.58), G (0.92, 0.22), and H (0.19, 0.81). Points C–F are clustered together, A and H are adjacent, and B and G are neighbors to each other. The sorting results of eight points are based on NSGA-II, and the improved version is shown in Fig. 1. The legends in the top-right corner record the order of each point.

The specific sorting process of the above example based on the improved NSGA-II is as follows. A and B are individuals corresponding to the minimum values of the two objective functions, so they are first moved into V_{in}. The distances from each remaining individual to individuals A and B are calculated, and the shorter distance is retained. Afterward, the distances of all individuals are compared, and the individual with the highest value is moved from V_{out} to V_{in}. In this example, D is the individual, followed by individuals A and B. Then, the distances between the remaining individuals and individuals in V_{in} are compared, including A, B, and D. The above steps are repeated until all individuals are moved into V_{in}.

If three individuals are selected and retained for the next iteration, in addition to individuals A and B, individual C will be selected by NSGA-II because of its higher crowding distance. Fig. 1 shows that individual D can better demonstrate the diversity of this front compared with individual C, because individual D is closer to the midpoint of individuals A and B. This is the reason why D has a higher priority than C in Fig. 1(b). In addition, when four individuals need to be reserved, the NSGA-II will select individual G following points A–C. However, individual G is close to point B, which is not a good choice. Based on the improved NSGA-II, points A, B, D, and F are selected, which demonstrates a more even distribution. According to the above experiments, the improved NSGA-II can demonstrate better diversity maintenance capability than the original version. The main reason is that NSGA-II only considers the distance between adjacent

Algorithm 1 Improved NSGA-II

Input: F (A set of objective function values for all individuals), POP (A set of individuals).

Output: SP (Superior subpopulation selected from all individuals), IP (Inferior subpopulation selected from all individuals), F' (the objective function values after reordering).

All individuals are assigned to the corresponding front $Front_i$ based on the dominance relationship.
Generate an empty vector S.
for each $Front_i$ **do**
    Generate vector V_{out} to record the index of all individuals on the front.
    Generate vector V_{in}.
    for k=1:m **do**
        $f_k^{\max}$=max(F_k)
        $f_k^{\min}$=min(F_k)
        L=$find(F_k == f_k^{\min})$
        Remove L from V_{out} into V_{in}.
    end for
    for $a = 1 : length(Front_i)$ **do**
        for $b = a : length(Front_i)$ **do**
            Calculate the relative distance between individuals a and b via Equation 10.
        end for
    end for
    for $c = 1 : length(Front_i) - m$ **do**
        Extract the distance between the individuals in V_{out} and the individuals in V_{in}.
        for $d = 1 : length(V_{\text{out}})$ **do**
            The minimum distance from individual d in V_{out} to all individuals in V_{in} is recorded as D_d.
        end for
        Find the individual L in V_{out} corresponding to the maximum value in D.
        Remove L from V_{out} into V_{in}.
    end for
    $S = [S; V_{\text{in}}]$
end for
The first $N/2$ individuals in S are classified as SP, and the $N/2$ individuals after SP are considered as IP, the objective function value corresponding to individuals in SP and IP is F'.
Return SP, IP, F'.

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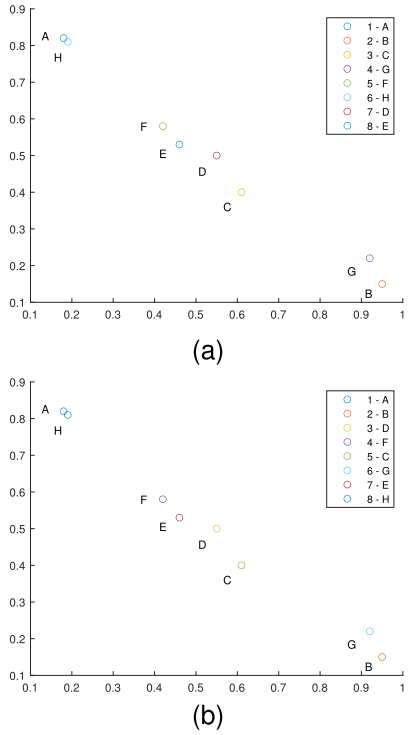


Fig. 1. Sorting results of different algorithms for the same individual. (a) NSGA-II. (b) Improved NSGA-II.

individuals, but the improved NSGA-II comprehensively considers the distance between all individuals, and the next reserved individual needs to maintain the maximum distance from the selected point.

C. Individual Update Strategies

Based on Idea 1 mentioned in the introduction, different individuals have distinct characteristics. Superior individuals are more suitable for the local search than those with poor fitness. Conversely, inferior individuals are more likely to search globally. Based on this, this article proposes different individual updating strategies for different individuals.

1) Global Search Strategy Based on Roulette Wheel Selection: Endmember extraction is a positive feedback process; that is, pixels that appear more frequently in the population are more likely to be defined as endmembers. Considering the positive feedback and discrete characteristics of endmember extraction, we modify the roulette wheel selection algorithm for global search. The algorithm contains three steps: 1) generate the pixel occurrence vector; 2) calculate the probability and cumulative probability of each pixel; and 3) generate an offspring. The specific steps of the global search are as follows, and its pseudocode is presented in Algorithm 2.

a) Generate pixel occurrence vector: To record the occurrence number of each pixel, a zero vector with the length of NP is generated. Each element in the vector represents the number of pixels that appear in the population. However, if the global search only considers the pixels that appear in the population, it is bound to become trapped in the local optimum. As such, the value of the pixels not appearing in the population is set as a constant related to the number of iterations. The value of each pixel in the vector can be obtained as follows:

$$O(i) = \begin{cases} \text{count}(i), & i \in R_1 \\ 1 - \dfrac{\text{iter}}{\text{max_iter}}, & i \in R_2 \end{cases} \tag{11}$$

where R_1 represents the set of pixels appearing in the population and R_2 is the set of other pixels in the image. iter indicates the number of current iterations, and max_iter signifies the total number of iterations. The linear expression in (11) can give pixels in R_2 an approximate value to pixels in R_1 at the prophase of updating. However, as the number of current iterations increases, the value for pixels that do not appear in the population gradually decreases. The value of pixels not in the population is zero in the last iteration.

b) Calculate the probability and cumulative probability of each pixel: After counting or calculating the vector value of each pixel in the current iterations, the probability of each pixel in the next iteration can be obtained by calculation. The specific equation is as follows:

$$p(i) = \frac{O(i)}{\sum_{i=1}^{NP} O(i)} \tag{12}$$

where $p(i)$ indicates the selection probability of the ith pixel in the next iteration.

The cumulative probability of each pixel is the sum of the selection probabilities of all pixels before the corresponding pixel. The specific equation is as follows:

$$q(i) = \sum_{j=1}^{i} p(j). \tag{13}$$

c) Generate an offspring: An offspring can be generated as steps 6–11 in Algorithm 2. A random number is first

Algorithm 2 Global Search

Input: $Temp_P$ (A population for global search).
Output: son (An offspring generated by global search).

1: Generate a zero vector O with the length NP.
2: Count the occurrences of each pixel in the $Temp_P$ via Equation 11.
3: Calculate the occurrences of each pixel not in the $Temp_P$ via Equation 11.
4: Calculate the probability of each pixel $p(i)$ via Equation 12.
5: Calculate the cumulative probability of each pixel $q(i)$ via Equation 13.
6: **for** $i = 1 : P$ **do**
7: $temp = q - rand$
8: Find the pixel L corresponding to the minimum positive value in $temp$.
9: $son(i) = L$
10: $q(L) = inf$
11: **end for**
12: **Return** son.

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generated and subtracted by the cumulative probability vector. The rand appearing in step 7 represents a random number ranging from 0 to 1. The pixel corresponding to the minimum positive value is selected as the first offspring element. As shown in step 10, the cumulative probability of the selected pixel is reset as an infinite value to avoid repeated selection. Eventually, steps 7–10 are repeated until P pixels are selected to generate a new offspring.

2) Local Search According to the Genetic Search: The main idea of local search is to recombine the genes (endmembers) of two parents to generate a new individual. The GA [53] is a classical heuristic algorithm with strong extensibility that is often used to optimize continuous problems. However, endmember extraction is a discrete combinatorial optimization problem. Therefore, a modified GA is proposed for endmember extraction in this section. To prevent some individuals with excellent endmembers from not being selected or the same parent combination from appearing in the next iteration, the individuals of the subpopulation are selected in a certain order. A sequence representing individual numbers is generated. Then, individuals corresponding to adjacent serial numbers are considered two parents to generate one offspring. For example, for the sequence [2 3 1 2], in this case, the second and third individuals are selected as the parents of the first offspring. Similarly, the first and second individuals are selected as the parents of the last offspring. Before recombination, duplicate endmembers in the two individuals are first removed. If the number of remaining endmembers exceeds $P + 1$, an offspring different from the parents may be generated by local search. The specific steps are shown in Algorithm 3. When the remaining endmembers are insufficient for generating a descendant dissimilar to the parents, the local search strategy will fail. In this case, the global search strategy will be mobilized to generate a new descendant instead of local search.

3) Offspring Complement Mechanism: Endmember extraction is an unordered combinatorial task. During offspring generation, there may be some individuals that contain the same endmembers as previous individuals with different orders. To avoid repeated calculations, a historical archive is built to store all the individuals generated in the previous iteration. The specific steps are shown in Algorithm 4. As shown in steps 1–4, the individuals generated in the current iterations are compared with all the individuals in the historical archive. If there are currently generated individuals that are the same as those in the historical archive, the global search algorithm is called to generate new individuals until N offsprings not appearing in the historical archive are generated. Finally, as in steps 5 and 6, the same individuals in the offspring are removed, leaving only N individuals that do not appear in the historical archive. Then, new individuals are generated by global search until N individuals are all distinct from the individuals in the historical archive.

Algorithm 3 Local Search

Input: $Temp_P$ (A population for local search)
Output: Son (Offspring generated by local search)

```
Generate a random sequence Order= randperm (length (Temp_P))
Order = [Order; Order(1)]
Son = ∅
for i = 1 : length(Temp_P) do
    Parent1 = Temp_P(Order(i))
    Parent2 = Temp_P(Order(i + 1))
    Parent = unique([Parent1; Parent2])
    if length(Parent) > P+1 then
        flag = 0
        while flag=0 || son=Parent1 || son=Parent2 do
            Select P elements from Parent to form a new individual son.
            flag = 1
        end while
    else
        son =Global search(Temp_P) //Algorithm 2
    end if
    Son = Son ∪ son
end for
Return Son
```

Algorithm 4 OCM

Input: HA_i (A historical archive in the i-th iteration), Son (All offspring generated in the i-th iteration)
Output: HA_{i+1} (The historical archive for the i+1-th iteration), Son' (Updated offspring after OCM)

```
while card(HA_i − Son) < N do
    son =Global search([SP, IP])//Algorithm 2
    Son' = Son ∪ son
end while
Son' = Son' − HA_i
HA_{i+1} = HA_i ∪ Son'
Return HA_{i+1}, Son'
```

D. RWS-GA

Combining the above strategies, the RWS-GA algorithm for endmember extraction is proposed. After initializing and evaluating all individuals, the population is divided into superior and inferior subpopulations based on the improved NSGA-II algorithm. Subsequently, global exploration and local exploitation are adopted in parallel for diverse subpopulations based on different evolutionary algorithms. In addition, the OCM strategy is adopted to guarantee enough new individuals to avoid wasting computing resources. In summary, parents and children compete together based on the NSGA-II, and only N excellent individuals are retained for the next iteration.

The pseudocode of the proposed method is displayed in Algorithm 5, and the process is graphically shown in Fig. 2.

IV. Experiments and Results

The method proposed in this article will be tested on three actual datasets. The three datasets contain three, four, and six endmembers. The comparison algorithms include traditional HEE algorithms, single-objective optimization algorithms,

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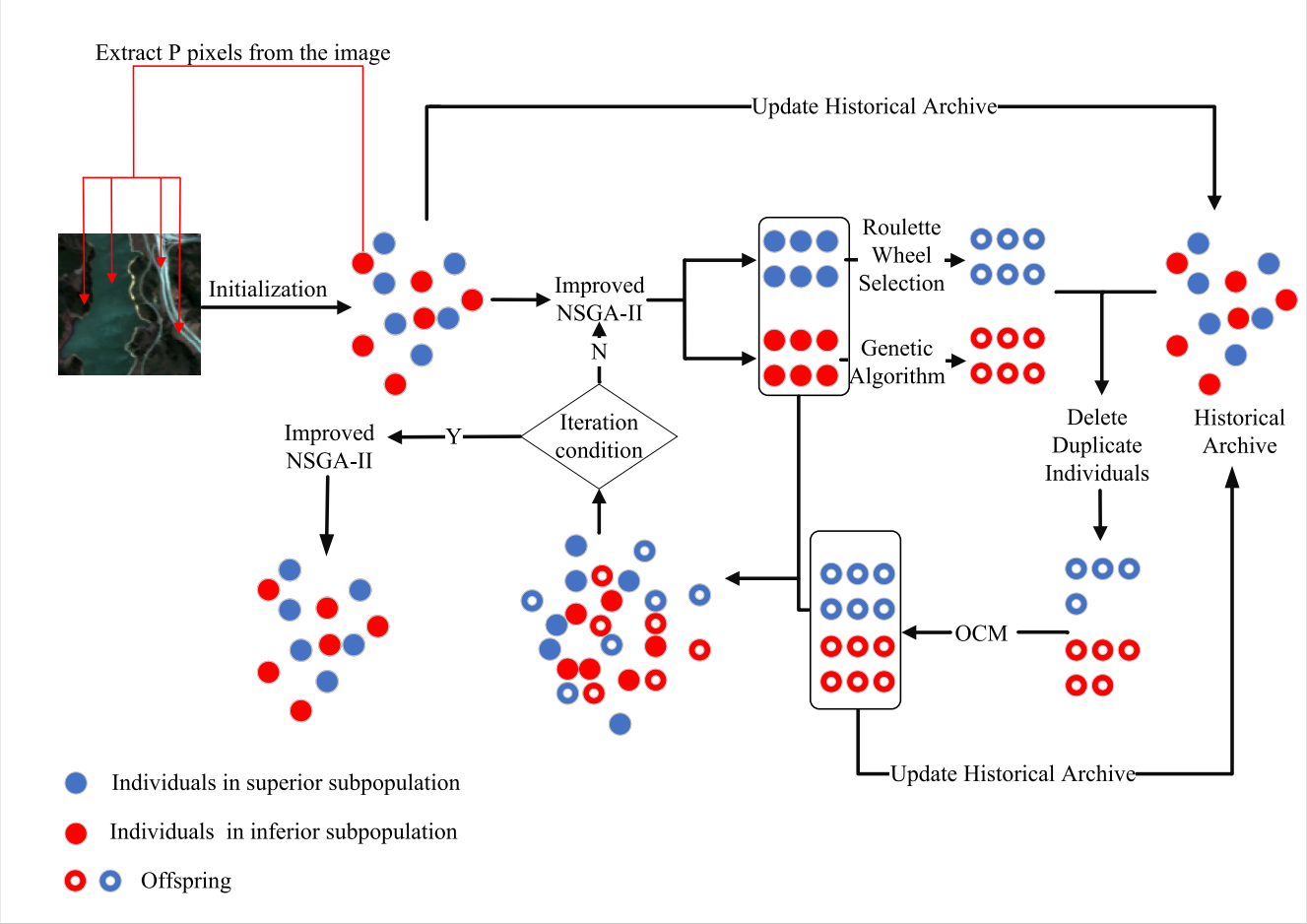


Fig. 2. Process of the proposed RWS-GA method.

and other multiobjective optimization algorithms, specifically vertex component analysis (VCA), N-FINDR, DPSO, MOAQPSO, MODPSO, and $(\mu + \lambda)$-MODE. With respect to the original author, MODPSO employs fully constrained least squares (FCLS) as the abundance estimation method. However, considering consumption time, other algorithms adopt NCLS to evaluate abundance. The numbers of individuals and iterations are set to 20 and 300, respectively. For the PSO variants, the random selection probability is set to 0.2. The scaling factor and crossover probability for the $(\mu+\lambda)$-MODE algorithm are [0.6, 0.8, 1.0] and [0.1, 0.2, 1.0], respectively.

Due to the lack of a real Pareto front for the HEE, it is difficult to evaluate the performance of algorithms. To compare various algorithms, the single-run results of all algorithms are first displayed together. In addition, the Pareto fronts of all multiobjective algorithms after five runs are compared to reflect the potential of the proposed method and other multiobjective algorithms. The C metric [54] is adopted to display the dominant relationship between the two multiobjective algorithms. The specific expression is as follows:

$$C(X', X'') = \frac{\left|\{u \in X''; \exists v \in X'; v \preceq u\}\right|}{|X''|} \tag{14}$$

where $v \preceq u$ indicates that individual v is dominant or equal to individual u. When the value of $C(X', X'')$ is equal to 1, the individuals in X'' are equal to or dominated by X'. When the value of $C(X', X'')$ is equal to 0, no one individual is dominated by the set X'. To fully show the relationship between the two sets, $C(X', X'')$ and $C(X', X'')$ both need to be considered, because there is no direct relationship between the above two values.

Although the exact Pareto fronts of the datasets are not available, the known spectrum signal of each ground object can be considered as evidence to verify the extraction results of various algorithms. The spectral angle distance (SAD) is adopted to evaluate the similarity between the extracted endmember and the known spectrum. The SAD calculation equation is as follows:

$$\mathrm{SAD}(x, \hat{x}) = \arccos\left(\frac{x^T \cdot \hat{x}}{\| x^T \| \cdot \| \hat{x} \|}\right) \tag{15}$$

where x indicates the spectral signal extracted by various methods and $\hat{x}$ is the known spectrum corresponding to x. A smaller SAD value indicates that a pixel closer to the reference spectrum is extracted as an endmember.

A. Samson Dataset

The original Samson dataset contains 952 × 952 pixels, a subset containing 95 × 95 pixels is used herein, and each pixel is recorded in 156 channels, covering wavelengths

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TABLE I
TIME CONSUMPTION OF VARIOUS ALGORITHMS ON THE SAMSON DATASET

	VCA	N-FINDR	DPSO	MOAQPSO	MODPSO	$(\mu+\lambda)$-MODE	RWS-GA
Time (s)	0.45	0.49	19.08	32.94	21.36	4250.04	2293.92

Algorithm 5 RWS-GA

Input: N (Number of individuals), P (Number of pixels in each individual), IMG (The input image).

Output: POP (A set of individuals), F (A set of objective function values for all individuals).

Initialize population $POP = \{pop_1, pop_2, \ldots, pop_N\}$, and historical archive $HA = POP$.
Evaluate the objective function value of all individuals $Fit = \{F_1, F_2, \ldots, F_N\}$.
for $i = 1 : max_iter$ **do**
  $[SP, IP, Fit]$=I-NSGA-II(POP, Fit) //**Algorithm 1**
  for $j = 1 : length(IP)$ **do**
    son =Global search(IP)//**Algorithm 2**
    $Son_G = Son_G \cup son$
  end for
  Son_L =Local search(SP)//**Algorithm 3**
  $Son = Son_G \cup Son_L$
  $[HA, Son] = OCM(HA, Son)$//**Algorithm 4**
  $POP = IP \cup SP \cup Son$
  Evaluate the objective function value of all individuals $Fit = \{F_1, F_2, \ldots F_{2N}\}$
end for
$[SP, IP, Fit]$=I-NSGA-II(POP, Fit) //**Algorithm 1**
$POP = IP \cup SP$
Return POP, Fit

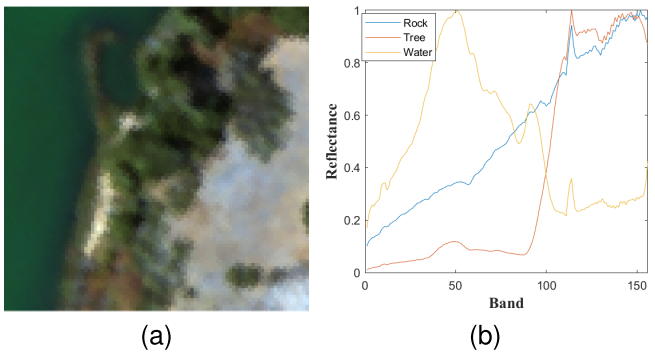


Fig. 3. Samson dataset. (a) False-color image of the Samson dataset. (b) Spectrum of each feature.

from 401 to 889 nm [55]. It starts from the (252, 332) pixel in the original image. The false-color image of the subset image is shown in Fig. 3(a). Its red, green, and blue channels correspond to the 94, 44, and 14 bands of the dataset, respectively. The subset image contains rock, tree, and water, and the corresponding spectral signatures of various ground objects are displayed in Fig. 3(b).

Fig. 4(a) provides the dominant relationship between the results of various methods. It can be observed in Fig. 4(a) that the RWS-GA algorithm is better than DPSO on both objective functions. In addition, the proposed method maintains a nondominant relationship with VCA, N-FINDR, and MOAQPSO. The above three algorithms exhibit better performance in terms of volume than the proposed algorithms but exhibit significantly different performance in terms of reconstruction errors. This is mainly because the above three algorithms are based on the volume composed of endmembers as the theoretical basis and objective function of the algorithm. DPSO and MOAQPSO only consider one objective function, so the obtained solution usually performs well in its corresponding objective function but performs poorly in other objective functions. In Fig. 4(a), the extraction results of the two single-objective algorithms conform to the above description. Based on the Pareto solutions of multiple objectives, the inverse of the volume and the reconstruction error of the image are indeed two conflicting objectives, which again proves the feasibility of the combination of multiobjective optimization and endmember extraction.

As the number of runs increases, the performance of algorithms can be further demonstrated, and better and fairer Pareto solutions can be detected. The Pareto fronts of multiobjective optimization algorithms after five runs are presented in Fig. 4(b). According to Fig. 4(a) and (b), the Pareto solutions of the proposed algorithm are generally superior to those of the other two multiobjective algorithms. According to formula (14), the C-metric values between the other two algorithms and the proposed algorithm can be calculated to further quantify the dominant relationship between the various algorithms. The Pareto solutions of RWS-GA are better than those of MODPSO and $(\mu + \lambda)$-MODE by 39% and 80%, respectively. In turn, the MODPSO and $(\mu + \lambda)$-MODE solutions dominate the solutions obtained by the proposed method by 7% and 30%, respectively. From the above indicators, the proposed method shows strong competitiveness in the Samson dataset.

The time consumption of various methods in the Samson dataset is given in Table I. In general, the running time is relatively reasonable according to its solution. The proposed method can obtain better results than the result of $(\mu + \lambda)$-MODE nearly half of the time. Endmember extraction is a typical combinatorial optimization problem. RMSE can describe the overall unmixing effect of the extracted endmembers, but it cannot show the extraction effect of a single endmember. Compared with RMSE, SAD can better demonstrate the extraction performance of various methods on a single endmember. Table II shows the comparison results between the extraction results of various methods and the ground truth. The star symbol in the table indicates that the relevant method does not extract the endmember of the corresponding feature. All methods perform well in the Samson dataset, except that VCA does not extract the corresponding endmember of water. This is mainly because of the few endmembers and diminutive image size, which reduces the requirement for optimization.

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TABLE II
MINIMUM SAD OF VARIOUS ALGORITHMS IN THE SAMSON DATASET

	VCA	N-FINDR	DPSO	MOAQPSO	MODPSO	$(\mu+\lambda)$-MODE	RWS-GA
Rock	0.0654	0.0404	0.0204	0.0404	0.0127	0.0084	0.0073
Tree	0.0219	0.0407	0.0331	0.0219	0.0175	0.0101	0.0219
Water	*	0.1296	0.0570	0.1304	0.0567	0.0297	0.0369

* :the endmember is not extracted by the method

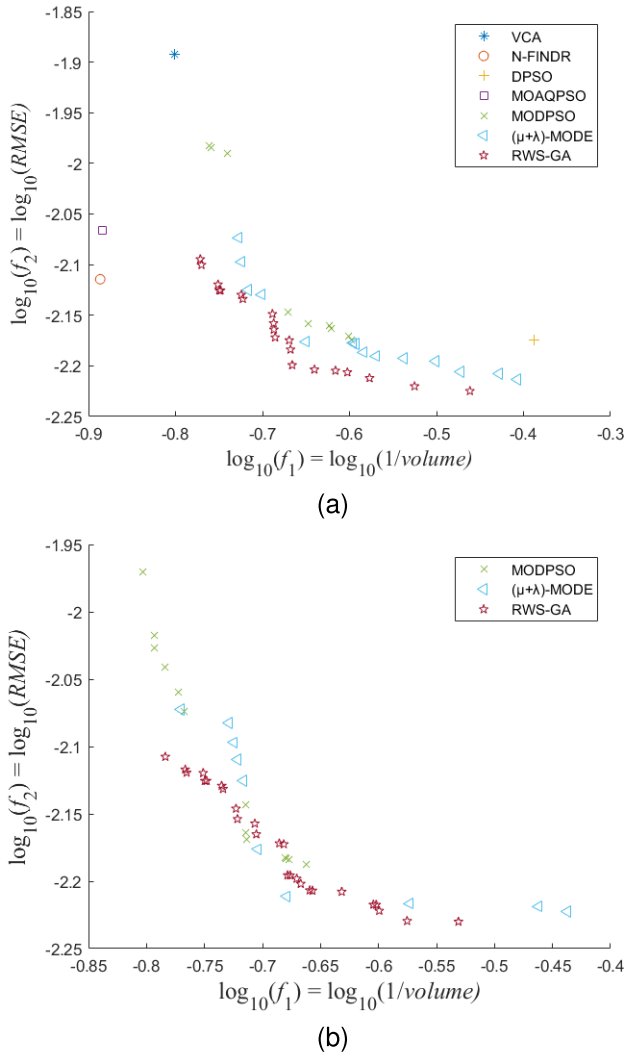


Fig. 4. Results of the Samson dataset. (a) Results obtained by all algorithms. (b) Pareto fronts of multiobjective algorithms running five times.

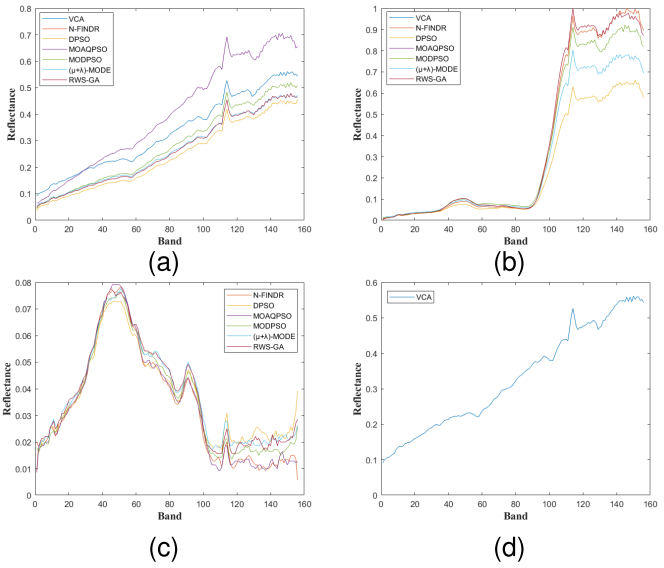


Fig. 5. Endmembers extraction by various algorithms for the Samson dataset. (a) Rock. (b) Tree. (c) Water. (d) Unmatched spectrum.

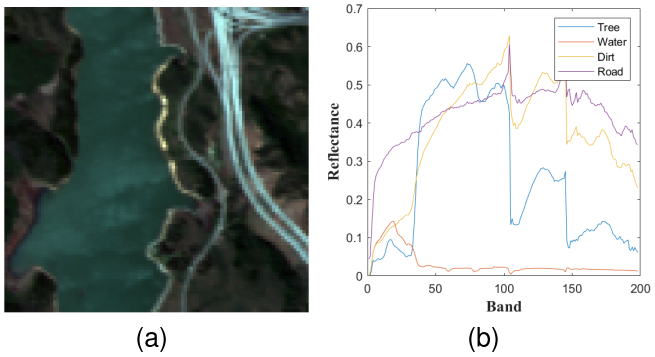


Fig. 6. Jasper Ridge dataset. (a) False-color image of the Jasper Ridge dataset. (b) Spectrum of each feature.

The extraction results of various features are shown in Fig. 5.

B. Jasper Ridge Dataset

The original image size of the Jasper Ridge dataset is 512 × 614, which contains 224 channels. There are 198 channels left after removing some channels vulnerable to water vapor and atmospheric conditions. This article adopts a 100 × 100 submap of Jasper Ridge, which contains four main features: trees, water, soil, and roads [56]. The false-color image and ground feature spectrum of the Jasper Ridge dataset are shown in Fig. 6.

The solutions of various methods are shown in Fig. 7(a), and the Pareto fronts obtained after five runs are displayed in Fig. 7(b). According to Fig. 7(a), the proposed method showed better performance than all algorithms on the reconstruction RMSE. RWS-GA dominates DPSO and has a nondominant relationship with other algorithms, including VCA, N-FINDR, and MOAQPSO. This phenomenon is analogous to the Samson dataset, which proves that the proposed algorithms have good stability. The performance of three multiobjective optimization algorithms on the Jasper dataset shows that the RWS-GA exhibits competitive advantages compared with the other two algorithms. In addition, further experiments shown in Fig. 7(b) reveal the overwhelming preponderance of the

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TABLE III
TIME CONSUMPTION OF VARIOUS ALGORITHMS ON THE JASPER RIDGE DATASET

	VCA	N-FINDR	DPSO	MOAQPSO	MODPSO	$(\mu+\lambda)$-MODE	RWS-GA
Time (s)	0.60	0.70	57.36	61.98	30.42	5070.36	2653.14

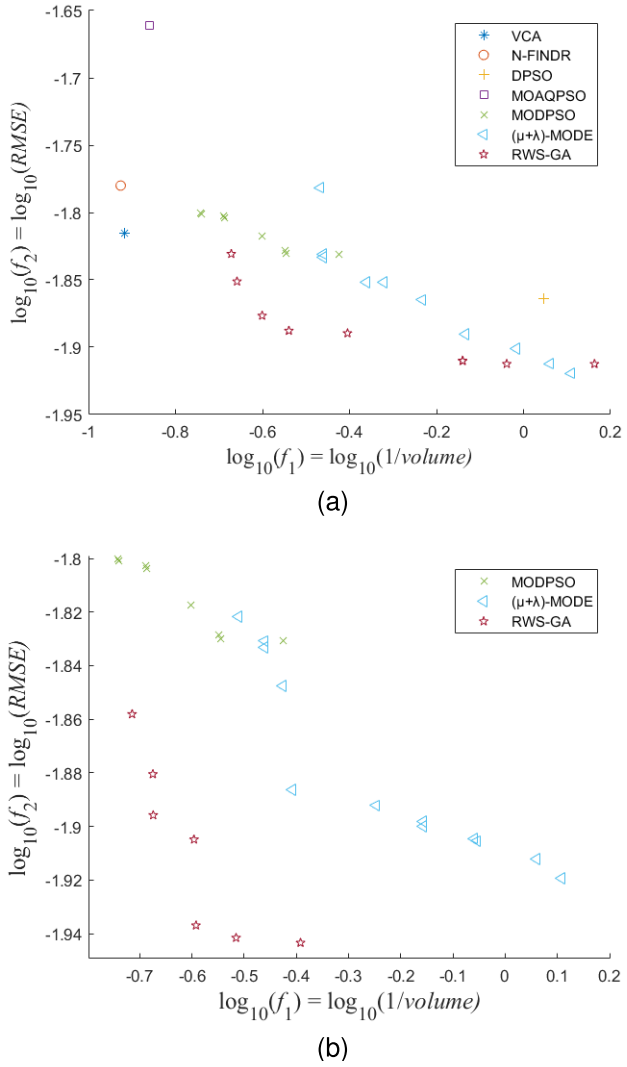


Fig. 7. Results of the Jasper Ridge dataset. (a) Results obtained by all algorithms. (b) Pareto fronts of multiobjective algorithms running five times.

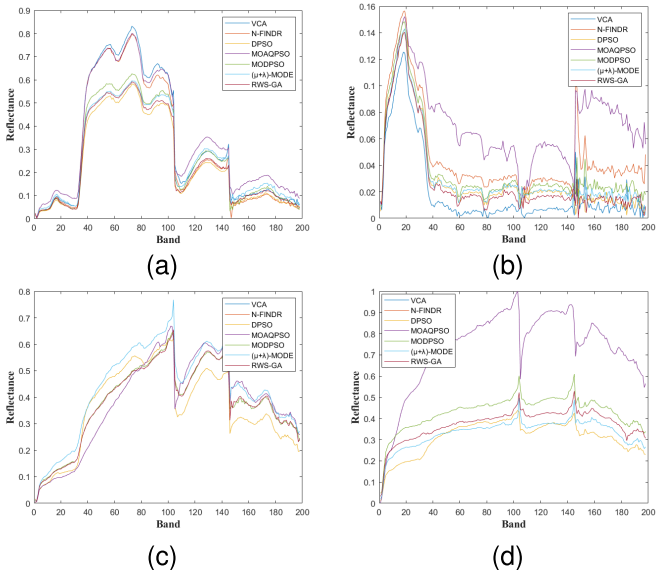


Fig. 8. Endmembers extraction by various algorithms for the Jasper Ridge dataset. (a) Tree. (b) Water. (c) Dirt. (d) Road.

proposed algorithm. The Pareto solutions obtained by the proposed method dominate 75% and 100% solutions from MODPSO and $(\mu+\lambda)$-MODE, respectively. In addition, there is no solution obtained by the other two algorithms dominating solutions from RWS-GA.

The time consumed by various methods on the Jasper Ridge dataset is also presented in Table III, which displays similar results, as shown in Table I; that is, the proposed algorithm obtains better solutions with a half-time of $(\mu+\lambda)$-MODE. The minimum SAD corresponding to various methods is shown in Table IV, and the extracted spectral signals of various ground objects are shown in Fig. 8. Combined with Figs. 7(b) and 8 and Table IV, although the solutions obtained by traditional methods and MOAQPSO show considerable volume, the minimum SAD values corresponding to the above three algorithms are generally higher than those of other endmember extraction algorithms. This is mainly because VCA, N-FINDR, and MOAQPSO have weak antinoise abilities; however, the remaining algorithms have good robustness and can resist the influence of noise.

C. Urban Dataset

Urban is also a common dataset in unmixing [57]. The dataset, with a size of 200 × 200, consists of 210 channels. After removing some water absorption bands, 162 bands describing the spectrum of each pixel are left for subsequent tests. There are six features in this image, including asphalt road, grass, trees, roof, metal, and dirt as shown in Fig. 9. Compared with the above two datasets, this dataset has more endmembers and a larger size, which means a high requirement for optimization algorithms.

PF is the most direct approach to demonstrate the effectiveness of algorithms. As previously set in the experiment, the single-endmember extraction results of various algorithms are shown in Fig. 10(a). This figure shows that single-objective optimization and traditional algorithms except DPSO have shown strong competitiveness in terms of volume, but most solutions obtained by the proposed algorithm are significantly better than the above three algorithms. Due to the randomness of evolutionary algorithms, most of the offspring generation process is related to random probability, so single-round experimental results cannot accurately reflect the comparative effect of multiobjective evolutionary algorithms. The experimental results of three multiobjective optimization algorithms running

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TABLE IV
MINIMUM SAD OF VARIOUS ALGORITHMS IN THE JASPER RIDGE DATASET

	VCA	N-FINDR	DPSO	MOAQPSO	MODPSO	$(\mu+\lambda)$-MODE	RWS-GA
Tree	0.1180	0.1559	0.0738	0.0603	0.0568	0.0195	0.0604
Water	0.2391	0.2453	0.0822	0.5432	0.1021	0.0690	0.0880
Dirt	0.1336	0.1336	0.1037	0.1336	0.0324	0.0221	0.0364
Road	0.1069	0.1069	0.1005	0.1069	0.0171	0.0260	0.0252

TABLE V
TIME CONSUMPTION OF VARIOUS ALGORITHMS IN THE URBAN DATASET

	VCA	N-FINDR	DPSO	MOAQPSO	MODPSO	$(\mu+\lambda)$-MODE	RWS-GA
Time (s)	6.19	8.84	498.61	734.35	286.42	57114.72	30819.62

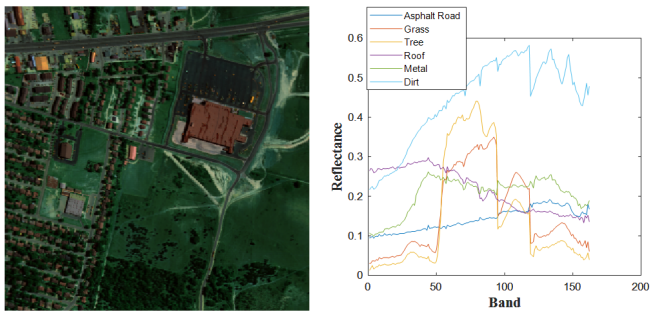


Fig. 9. Urban dataset. (a) False-color image of the Urban dataset. (b) Spectrum of each feature.

five times on the Urban dataset are shown in Fig. 10(b). In this figure, the proposed method dominates the other two multiobjective optimization algorithms, which indicates that the proposed algorithm also has application value in large-scale images in complex scenarios. In Fig. 10, MODPSO and PSO have a lower extraction effect on the Urban dataset compared with the above two datasets. This is because Urban has a larger search scale and decision number (endmembers) compared with the two datasets mentioned above. The two PSO-based discrete algorithms mentioned above are prone to falling into local optima due to their offspring generation strategy. However, the proposed discrete algorithm in this article can still maintain excellent competitiveness in the Urban dataset while maintaining the physical meaning of the extracted endmembers.

The time consumed by various methods on the Urban dataset is also given in Table V. The extraction situations of each method for features are illustrated in Fig. 11, and the corresponding minimum SAD is shown in Table VI. It can be noted that multiobjective optimization algorithms can not only provide multiple candidate solutions for decision-makers to choose from but also have some resistance to noise and outliers. Combining Fig. 11 and Table VI, traditional algorithms and single-objective optimization algorithms have difficulty extracting all ground features. The extraction algorithms with reconstruction error as the objective function have higher robustness than the volume-based algorithms. Moreover, the combination of the two objective functions can greatly improve the robustness of the algorithm.

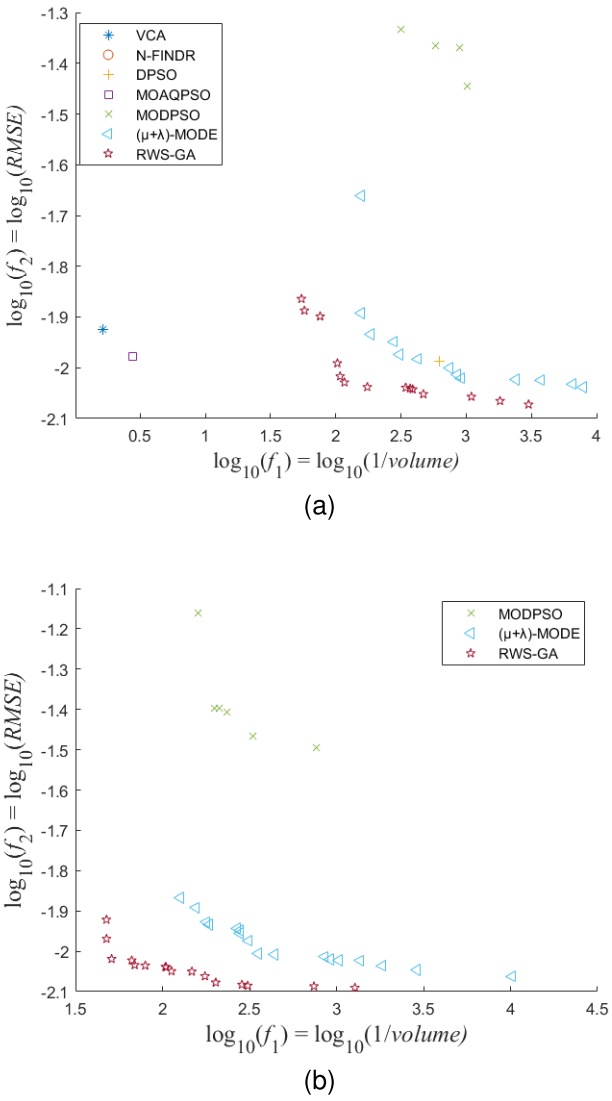


Fig. 10. Results of the Urban dataset. (a) Results obtained by all algorithms. (b) Pareto fronts of multiobjective algorithms running five times.

D. Ablation Experiments

In this section, the contribution of each strategy, including the improved NSGA-II, population segmentation strategy, parent-partner strategy, and historical archive strategy, is displayed. For the above strategies, five variants are designed and tested on Jasper Ridge datasets running five times.

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TABLE VI
MINIMUM SAD OF VARIOUS ALGORITHMS IN THE URBAN DATASET

	VCA	N-FINDR	DPSO	MOAQPSO	MODPSO	(μ+λ)-MODE	RWS-GA
Road	*	*	0.0320	0.1513	0.0377	0.0317	0.0406
Grass	0.1160	0.1160	0.1567	0.1160	0.0326	0.0256	0.0508
Tree	0.1614	0.1227	0.0817	0.1227	0.0414	0.0692	0.1148
Roof	0.1740	0.1740	0.0806	*	0.0469	0.0330	2.11E-8
Metal	*	0.3065	*	0.2077	0.0618	0.0301	0.0381
Dirt	*	*	0.0548	*	0.0477	0.0181	0.0719

* :the endmember is not extracted by the method

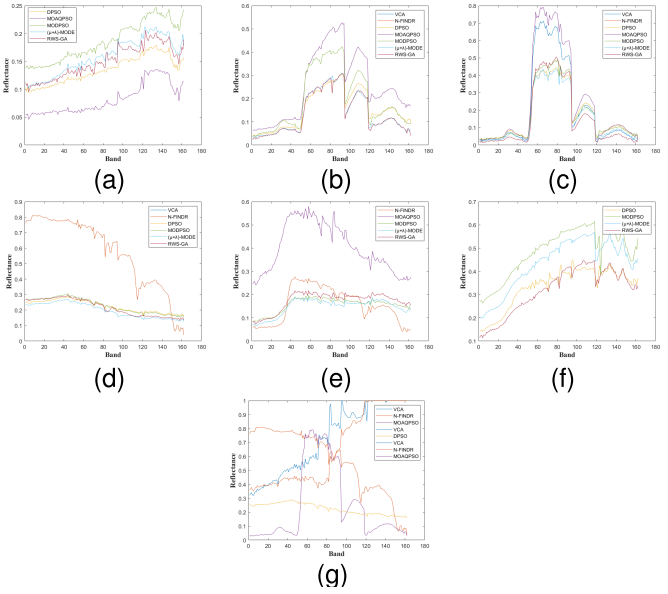


Fig. 11. Endmembers extraction by various algorithms for the Urban dataset. (a) Road. (b) Grass. (c) Tree. (d) Roof. (e) Metal. (f) Dirt. (g) Unmatched spectra.

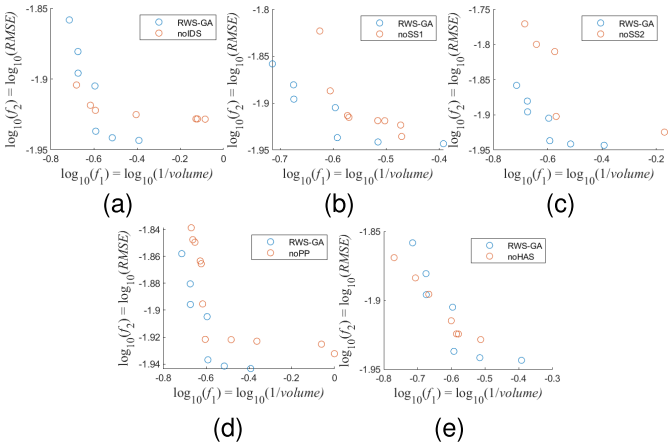


Fig. 12. Results between the proposed algorithm and other variants in the Jasper Ridge dataset. (a) noIDS. (b) noSS1. (c) noSS2. (d) noPP. (e) noHAS.

TABLE VII
C-METRIC VALUES OF THE PROPOSED ALGORITHM AND OTHER VARIANTS IN THE JASPER RIDGE DATASET

Indices \ Variants	noIDS	noSS1	noSS2	noPP	noHAS
C (proposed, Variant)	0.571	1	1	0.909	0.571
C (Variant, proposed)	0.429	0	0	0.143	0.429

1) *noIDS:* Variant that adopts the original NSGA-II instead of the improved NSGA-II.
2) *noSS1:* Variant that has no population segmentation strategy, and all individuals are updated with the modified GA.
3) *noSS2:* Variant that has no population segmentation strategy, and all individuals are updated with the roulette selection strategy.
4) *noPP:* Variant that applies a population segmentation strategy. However, there is no parent-partner strategy, and all parents are randomly selected in the GA.
5) *noHAS:* Variant without a historical archive that records previous individuals; the offspring are only compared with the current population.

The Pareto fronts of the proposed algorithm and various variants are shown in Fig. 12. The C-metric values between the proposed algorithm and variants are listed in Table VII. It can be seen in Fig. 12(b)–(d) that the RWS-GA algorithm can obtain better Pareto solutions than variants, including noSS1, noSS2, and noPP. As shown in these figures, it is significant to divide individuals into diverse populations and adopt different updating strategies for different subpopulations. The parent-partner strategy in the proposed algorithm also plays an important role in improving the algorithm performance. While it is not evident in Fig. 12(a) and (e) that RWS-GA has obvious advantages over noIDS and noHAS, Table VII shows that the proposed algorithm is still superior to the above two variants.

E. Discussion

Experiments on three real-world datasets show the potential of multiobjective optimization in endmember extraction. As shown in Tables II–VI, traditional methods and single-objective optimization algorithms often have a larger SAD value than multiobjective optimization algorithms and tend to consider mixed pixels or outliers as endmembers. However, the multiobjective algorithms showcase stronger antinoise ability and better endmember identification, which also reflects the advantages of multiobjective optimization in the HEE. Combined with Figs. 4(a), 7(a), and 10(a), the proposed algorithm is often weaker in volume than algorithms, including N-FINDR, VCA, and MOAQPSO but exhibits strong competitiveness in reconstruction errors. The above phenomenon is mainly because these three algorithms are designed based on the volume of the endmember set, so they exhibit opposite performances in terms of volume and reconstruction error. Figs. 4(b), 7(b), and 10(b) show that the proposed algorithm dominates the solutions of MODPSO and (μ+λ)-MODE, which once again proves the practicality of the

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two ideas proposed in this article. Moreover, with the increase in the number of endmembers and image size, the superiority of the proposed algorithm over other algorithms is gradually enhanced.

By comparing the proposed algorithm with five variants in the Pareto fronts and C-metric values, the effectiveness of various strategies proposed in this article is verified, including the improved sorting algorithm, three individual updating strategies, and parent pairing strategies. Among them, the population segmentation strategy and different update strategies for different populations contribute significantly to the results, which once again verifies the rationality of the two ideas proposed in the introduction section.

V. Conclusion

In this article, an RWS-GA for HEE was proposed. In the RWS-GA, to solve the multiobjective optimization problem in the HEE, the optimal solutions were approximated in a multistrategy updating manner. By further exploring the combination of endmember extraction and discrete optimization, two opinions are proposed. Based on the first opinion, an improved NSGA-II algorithm is first proposed for sorting individuals, which divides them into the superior subpopulation and inferior subpopulation. According to the second opinion, two evolutionary algorithms are modified for updating diverse populations in parallel. Among them, the roulette wheel selection algorithm is adopted and amended for the global search based on the inferior subpopulation. For the superior subpopulation, a modified GA is developed for local search, in which the parent-partner strategy is designed to guarantee that each individual participates in the offspring generation. In addition, calculating reconstruction errors consumes a significant amount of time. To avoid duplicate calculations, a historical archive is established to record the individuals generated in previous iterations. Ultimately, experiments on three remote-sensing datasets verify that the proposed algorithm is applicable for endmember extraction and superior to state-of-the-art comparison algorithms.

Although the proposed algorithm demonstrated strong competitiveness on the HEE, there are still some drawbacks that need further improvement, and there are some interesting works worth further research. As mentioned in this article, calculating reconstruction errors is a significant time-consuming task. Reducing evaluation time is a significant research work. Accelerating the convergence speed of the algorithm and designing a simpler and faster model to replace the original evaluation model are important means to solve the above problems. Moreover, hardware acceleration can also be adopted to speed up operation efficiency, which employs hardware modules to replace software algorithms to make full use of the inherent fast characteristics of the hardware. In addition to those discussed above, we attempt to apply multiobjective optimization algorithms to other problems involving hyperspectral unmixing, such as extracting endmember bundles and abundance estimation.

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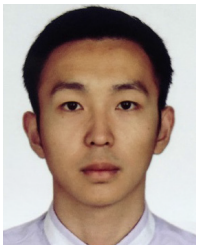
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